Bio 303 Quiz# 02
Collected by Zabi Attariya



What is the general mechanism of an enzyme?

Select the correct option

- ○ It acts by increasing the pH
- ○ It acts by decreasing the pH
- ○ It acts by increasing the activation energy
- ◉ It acts by reducing the activation energy

Click to Save Answer &

Question # 2 of 8 (Start time: 07:24:37 AM, 23 August 2022) To

Kinases belong to _____ group of enzymes.

Select the correct option

- ○ Isomerases
- ○ Oxidoreductases
- ○ Ligases
- ◉ Transferases

Click to Save Answer & Move to Next Q

Question # 3 of 8 (Start time: 07:25:23 AM, 23 August 2022)

Total Marks

Which one of the following reactions is catalyzed by Phosphotransfrases?

Select the correct option

- ○ Non-hydrolytic removal of a group from a substrate
- ○ Change of molecular form of substrate
- ◉ Transfer of specific group from one substrate to another
- ○ Hydrolysis of substrate

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:22 AM, 23 August 2022

Question # 4 of 8 (Start time: 07:26:56 AM, 23 August 2022)

Total Marks: 1

The transition state of a catalyzed reaction is

Select the correct option

- (●) higher in energy than that of an uncatalyzed reaction
- ( ) lower in energy than that of an uncatalyzed reaction
- ( ) a highly-populated intermediate on the reaction pathway
- ( ) lower in energy than the reaction substrate

Click to Save Answer & Move to Next Question

The transtion state of a catalyzed reaction is lower in energy than that of the uncatalyzed reaction because of enthalpic interactions between the enzyme and the transition state because of favorable interactions with the substrate_ because of a smaller delta $ between the [ES] and [ES"] complex because enzymes only work at low temperatures:

BIO303 - Biochemistry II (2nd quiz) Quiz Start Time: 07:22 AM, 23 August 2022

Question # 5 of 8 (Start time: 07:28:31 AM, 23 August 2022) Total Marks: 1

Decarboxylases belong to _____ group of enzymes.

Select the correct option

○	Oxidoreductases
○	Ligases
◉	Lyases
○	Isomerases

Click to Save Answer & Move to Next Question

Question # 6 of 8 (Start time: 07:29:44 AM, 23 August 2022)

A complex organic molecule or metallo-organic compound it is termed a _____

Select the correct option

- ○ Cofactor
- ◉ Coenzyme
- ○ Holo enzyme
- ○ Apoenzyme

Click to Save Answer & Move to Nex

Question # 7 of 8 (Start time: 07:30:28 AM, 23 August 2022) Total M

Digestive enzymes belong to _____ group of enzymes.

Select the correct option

- ○ Ligases
- ◉ Hydrolases
- ○ Isomerases
- ○ Oxidoreductases

Click to Save Answer & Move to Next Questio

Question # 8 of 8 (Start time: 07:31:04 AM, 23 August 2022)

RNAs make up ____ of the total RNA in the cell.

Select the correct option

- ○ 20%
- ◉ 80%
- ○ 5%
- ○ 50%

Click to Save Answer & Mov

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time:**

Question # 1 of 8 (Start time: 07:22:36 AM, 23 August 2022)

What is the general mechanism of an enzyme?

Select the correct option

- ○ It acts by increasing the pH
- ○ It acts by decreasing the pH
- ○ It acts by increasing the activation energy
- ● It acts by reducing the activation energy

Click to Save Answer &

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:32 AM, 23 August 2022

Question #1 of 8 (Start time: 07:32:33 AM, 23 August 2022)

Total Marks: 1

Induced fit serves to bring specific _____on the enzyme into the proper position to catalyze the reaction

Select the correct option

- (●) Functional groups
- ( ) Sulphide residues
- ( ) Enzyme
- ( ) Amino Acid

Click to Save Answer & Move to Next Question

Question # 2 of 8 (Start time: 07:33:29 AM, 23 August 2022) **Total Marks: 1**

Acceptor arm of tRNA is made up of unpaired sequences of _____ at the 3'end.

Select the correct option

- ○ Adenine-adenine-cytosine
- ◉ cytosine-cytosine-adenine
- ○ Adenine-adenine-Adenine
- ○ Adenine-cytosine-adenine

Click to Save Answer & Move to Next Question

Question # 3 of 8 (Start time: 07:34:08 AM, 23 August 2022)
Total Marks: 1

During an enzyme-catalyzed reaction change in conformation when the substrate binds is induced by rearrangements of_____

Select the correct option

- ○ Hydrogen bond
- ◉ Covalent bond
- ○ Disulphide bond
- ○ Ionic bond

Click to Save Answer & Move to Next Question

Question # 4 of 8 (Start time: 07:35:39 AM, 23 August 2022) **Total Marks: 1**

A _____binds at a site on the protein called the binding site

Select the correct option

○	Prostetic group
○	Enzyme
◉	Ligand
○	CO enzyme

Click to Save Answer & Move to Next Question

Question # 5 of 8 (Start time: 07:36:25 AM, 23 August 2022) **Total Marks: 1**

A catalyst increases the rate of a reaction by _____

Select the correct option

- ○ decreasing the concentration of the reactant(s)
- ○ increasing the concentration of reactant(s)
- ◉ decreasing the activation energy of the overall reaction
- ○ increasing the activation energy of the overall reaction

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:32 AM, 23 August 2022

Question # 6 of 8 (Start time: 07:38:03 AM, 23 August 2022)

Total Marks: 1

Which one of the following reactions is catalyzed by Phosphotransfrases?

Select the correct option

○	Non-hydrolytic removal of a group from a substrate
○	Change of molecular form of substrate
◉	Transfer of specific group from one substrate to another
○	Hydrolysis of substrate

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:32 AM, 23 August 2022

Question # 7 of 8 (Start time: 07:39:38 AM, 23 August 2022)

Total Marks: 1

Removal of phosphoryl groups is catalyzed by_____

Select the correct option

- (●) Protein phosphatases
- ( ) Dinitrogenase reductase
- ( ) Diphteria toxin and cholera toxin
- ( ) Protein kinases

Click to Save Answer & Move to Next Question

Question # 8 of 8 (Start time: 07:41:04 AM, 23 August 2022) T

What is the function of phosphorylase?

Select the correct option

- ○ Use H2O2 as the electron acceptor
- ○ Transfer a carboxylate group
- ◉ Transfer amino group
- ○ Transfer inorganic phosphate

Click to Save Answer & Move to Next

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:44 AM, 23 August 2022

Question #1 of 8 (Start time: 07:44:49 AM, 23 August 2022)

Total Marks: 1

Which one of the following acts as substrate for peptidases?

Select the correct option

- (●) Protein
- ( ) Lipid
- ( ) Carbohydrate
- ( ) Nucleic acid

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz) Quiz Start Time: 07:44 AM, 23 August 2022

Question # 2 of 8 (Start time: 07:46:11 AM, 23 August 2022) Total Marks: 1

The free energy of the ground state of product is _____than that of substrate

Select the correct option

- (●) Lower
- ( ) Greater
- ( ) Different
- ( ) Equal

Click to Save Answer & Move to Next Question

Question # 3 of 8 (Start time: 07:46:45 AM, 23 August 2022) **Total Marks: 1**

Citric acid Synthetase belong to _____ group of enzymes.

Select the correct option

- (●) Ligases
- ( ) Lyases
- ( ) Oxidoreductases
- ( ) Isomerases

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:44 AM, 23 August 2022

Question # 4 of 8 (Start time: 07:47:32 AM, 23 August 2022)

Total Marks: 1

Which of the following enzymes catalyze oxidation-reduction reaction?

Select the correct option

◉	Oxidoreductases
○	Lyases
○	Transferases
○	Hydrolases

Click to Save Answer & Move to Next Question

Question # 5 of 8 (Start time: 07:47:54 AM, 23 August 2022) **Total Marks: 1**

Which of the following enzymes catalyze the transfer of hydrogen atoms from one substrate to another?

Select the correct option

- (●) Transferases
- ( ) Lyases
- ( ) Oxidoreductases
- ( ) Hydrolases

Click to Save Answer & Move to Next Question

Question # 6 of 8 (Start time: 07:48:14 AM, 23 August 2022) **Total Marks: 1**

The starting point for either the forward or the reverse reaction is called the _____

Select the correct option

- (●) Ground state
- ( ) Activation stage
- ( ) lag state
- ( ) Log state

Click to Save Answer & Move to Next Question

Question # 7 of 8 (Start time: 07:48:34 AM, 23 August 2022) **Total Marks: 1**

The surface of the active site of enzyme is lined with _____ residues with side chains that complement and bind the substrate and catalyze its chemical transformation.

Select the correct option

- ○ Triglycerides
- ◉ amino acid
- ○ fatty acids
- ○ monosaccharides

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 07:44 AM, 23 August 2022

Question # 8 of 8 (Start time: 07:49:43 AM, 23 August 2022)

Total Marks: 1

Aldolases belong to _____ group of enzymes

Select the correct option

- (●) Lyases
- ( ) Ligases
- ( ) Oxidoreductases
- ( ) Isomerases

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time: 08:20 PM, 23 August 2022**

Question # 1 of 8 (Start time: 08:20:46 PM, 23 August 2022) **Total Marks:** 1

When a substrate combines with an enzyme, it induces a change in the_____ conformation

Select the correct option

- Product's
- enzyme's
- Subtrate's
- All given

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 08:20 PM, 23 August 2022

Question # 2 of 8 (Start time: 08:23:18 PM, 23 August 2022)

Total Marks: 1

Hydroperoxidases belong to _____ group of enzymes.

Select the correct option

- ○ Oxidoreductases
- ○ Lyases
- ○ Ligases
- ○ Isomerases

Click to Save Answer & Move to Next Question

Question # 3 of 8 (Start time: 08:25:13 PM, 23 August 2022) **Total Marks:**

Which one of the following enzymes catalyze the conversion of 2-Phosphoglycerate to Phosphoenolpyruvate?

Select the correct option

- ○ Phospholipase
- ○ Enolase
- ◉ Hydrolase
- ○ Peptidase

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 08:20 PM, 23 August 2022

Question # 4 of 8 (Start time: 08:27:08 PM, 23 August 2022)

Total Marks: 1

All tRNA molecules contain ___main arms or loops.

Select the correct option

- ○ 3
- ○ 4
- ○ 5
- ○ 2

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz) Quiz Start Time: 08:20 PM, 23 August 2022

Question # 5 of 8 (Start time: 08:29:38 PM, 23 August 2022) Total Marks: 1

Which of the following enzymes catalyze the from non-hydrolytic removal of a group from a substrate?

Select the correct option

- Isomerases
- Lyases
- Oxidoreductases
- Transferases

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time: 08:20 PM, 23 August 2022**

Question # 7 of 8 (Start time: 08:32:13 PM, 23 August 2022) **Total Marks:** 1

The surface of the active site of enzyme is lined with _____ residues with side chains that complement and bind the substrate and catalyze its chemical transformation.

Select the correct option

- ○ monosaccharides
- ○ fatty acids
- ○ amino acid ✓
- ○ Triglycerides

Click to Save Answer & Move to Next Question

Question # 1 of 8 (Start time: 08:39:17 PM, 23 August 2022)

During an enzyme-catalyzed reaction change in conformation when the substrate binds is induced by rearrangements of

Select the correct option

- Hydrogen bond
- Ionic bond
- Covalent bond
- Disulphide bond

Click to Save Answer & Move

Question # 3 of 8 (Start time: 08:42:59 PM, 23 August 2022)

The equilibrium between S and P reflects the difference in the free energies of their _____

Select the correct option

- Free state
- Ground states
- Initial state
- Binding state

Click to Save

Question # 4 of 8 (Start time: 08:44:04 PM, 23 August 2022)

The molecule which acts directly on an enzyme to lower its catalytic rate is _____

Select the correct option

- Inhibitor
- Repressor
- Regulator
- Modulator

Question # 5 of 8 (Start time: 08:45:52 PM, 23 August 2022)

The progressive chemical changes of a reaction are expressed by _____

Select the correct option

○	Right angle
○	Vericle axis
✓	Horizontal axis
○	All given

Question # 6 of 8 (Start time: 08:47:04 PM, 23 August 2022)

Aldolases belong to _____ group of enzymes.

Select the correct option

●	Lyases
○	Ligases
○	Isomerases
○	Oxidoreductases

Question # 7 of 8 (Start time: 08:48:27 PM, 23 August 2022)

Which step of a reaction is the rate-determining step?

Select the correct option

○	the fastest step
○	the slowest step
○	the first step
○	the last step of the reaction mechanism

Enzymes

Select the correct option

- Do not require activation energy
- Increase requirement of activation energy
- Do not change requirement of activation energy
- Decrease requirement of activation energy

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 08:20 PM, 23 August 2022

Question # 8 of 8 (Start time: 08:34:24 PM, 23 August 2022)

Total Marks: 1

The difference between the energy levels of the ground state and the transition state is the _____

Select the correct option

- (●) Activation energy
- ( ) Free Energy
- ( ) Transition energy
- ( ) Ground state energy

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time: 08:56 PM, 23 August 2022**

Question # 2 of 8 (Start time: 08:58:07 PM, 23 August 2022) **Total Marks:** 1

Which one of the following enzymes catalyze the removal of H_2O leaving behind double bonds?

Select the correct option

- ○ Phospholipase
- ◉ Lyase
- ○ Peptidase
- ○ Hydrolase

Click to Save Answer & Move to Next Question

Question # 4 of 8 (Start time: 09:01:43 PM, 23 August 2022) **Total Marks:**

Binding site of a protein is complementary to the ligand in_____

Select the correct option

- ◯ All given
- ◯ Shape
- ◯ Size
- ◯ Charge

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 09:30 PM, 23 August 2022

Question # 1 of 8 (Start time: 09:30:38 PM, 23 August 2022)

Total Marks: 1

The _____ complex existence is central to the action of enzymes

Select the correct option

- PE
- SP
- ES
- EP

ES

Click to Save Answer & Move to Next Question

BIO303 – Biochemistry II (2nd quiz)

Question # 2 of 8 (Start time: 09:31:45 PM, 23 August 2022)

Which of the statements regarding enzymes is false?

Select the correct option

○	Enzymes are specific
○	Enzymes are proteins that function as catalysts
○	Enzymes provide activation energy for reactions
○	Enzyme activity can be regulated

BIO303 - Biochemistry II (2nd quiz)

Question # 3 of 8 (Start time: 09:32:33 PM, 23 August 2022)

Which of the following enzymes catalyze the transfer of hydrogen atoms from one substrate to another?

Select the correct option

- ○ Transferases
- ○ Lyases
- ○ Oxidoreductases
- ○ Hydrolases

BIO303 - Biochemistry II (2nd quiz) Quiz Start Ti

Question # 4 of 8 (Start time: 09:33:53 PM, 23 August 2022)

_____ is a complex organic molecule or metallo-organic compound

Select the correct option

- Coenzyme
- All given
- Apoenzyme
- Holoenzyme

Click to Save Answ

Question # 5 of 8 (Start time: 09:34:39 PM, 23 August 2022)

47. The rate of an enzyme-catalyzed reaction increases with _____until a maximal velocity (Vmax) is reached.

Select the correct option

- ○ Product concentration
- ○ Substrate concentration
- ○ Ph
- ○ Temperature

Click to Save

BIO303 - Biochemistry II (2nd quiz) Quiz Start Ti

Question # 6 of 8 (Start time: 09:35:51 PM, 23 August 2022)

Digestive enzymes belong to _____ group of enzymes.

Select the correct option

- ○ Isomerases
- ○ Ligases
- ○ Hydrolases
- ○ Oxidoreductases

Click to Save Answ

Question # 7 of 8 (Start time: 09:37:00 PM, 23 August 2022)

The enzyme which forms the peptide bond is known as _____

Select the correct option

○	Peptidase
○	Peptidyle transferase
○	Carbonic anhydrase
○	carbohydrase

Click to Save

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time**

Question # 8 of 8 (Start time: 09:37:56 PM, 23 August 2022)

Oxygenases belong to _____ group of enzymes.

Select the correct option

- Ligases
- Lyases
- Oxidoreductases
- Isomerases

Click to Save Answer &

Question # 1 of 8 (Start time: 10:05:25 PM, 23 August 2022)

The starting point for either the forward or the reverse reaction is called the _____

Select the correct option

○	Activation stage
○	lag state
○	log state
○	Ground state

Question # 2 of 8 (Start time: 10:06:21 PM, 23 August 2022)

Which one of the following enzymes catalyze the conversion of fumarate to malate?

Select the correct option

○	Fumarase
○	Hydrolase
○	Enolase
○	Peptidase

Click to

Which of the following enzymes catalyze the hydrolysis of substrate?

Select the correct option

○	Transferases
○	Hydrolases
○	Oxidoreductases
○	Lyases

Click to

Question # 4 of 8 (Start time: 10:10:28 PM, 23 August 2022)

The force driving the system toward _____ is defined as standard free-energy change.

Select the correct option

- Equilibrium
- All of given
- Reactants
- Products

Click to Save

Question # 5 of 8 (Start time: 10:31:43 PM, 23 August 2022)

_____ is a signal triggering molecule binding to a site on target protein

Select the correct option

- ○ Substrate
- ○ Ligand
- ○ Active site
- ○ Enzyme

Click to Save

BIO303 – Biochemistry II (2nd quiz)

Question # 6 of 8 (Start time: 10:12:46 PM, 23 August 2022)

Decarboxylases belong to _____ group of enzymes.

Select the correct option

- ○ Oxidoreductases
- ○ Lyases
- ○ Ligases
- ○ Isomerases

Digestive enzymes belong to _____ group of enzymes.

Select the correct option

- Oxidoreductases
- Hydrolases
- Isomerases
- Ligases

Question # 8 of 8 (Start time: 10:15:17 PM, 23 August 2023)

The minimum amount of energy needed to start a reaction is called the _____

Select the correct option

- ○ reaction mechanism energy
- ○ activation energy
- ○ entropy of reaction
- ○ energy of reaction

Click to Save

Question # 1 of 8 (Start time: 10:17:21 PM, 23 August 2022)

Zymogen is

Select the correct option

- Enzyme Poison
- Enzyme precursor
- Enzyme inhibitor
- Enzyme modulator

Click to S

BIO303 - Biochemistry II (2nd quiz) Quiz Start Time: 1

Question # 2 of 8 (Start time: 10:18:40 PM, 23 August 2022)

Mobile genetic elements can be transferred from one DNA portion to another. The enzyme carrying out this is _____

Select the correct option

- Ligase
- Transposase
- Transcriptase
- Endonuclease

Click to Save Answer & M

Question # 3 of 8 (Start time: 10:20:18 PM, 23 August 2022)

The progressive chemical changes of a reaction are expressed by _____

Select the correct option

○	Right angle
○	Horizontal axis
○	All given
○	Vericle axis

Click

BIO303 - Biochemistry II (2nd quiz) Quiz Start Tir

Question # 4 of 8 (Start time: 10:21:23 PM, 23 August 2022)

Which one of the following reactions is catalyzed by synthetase enzyme?

Select the correct option

- Change of molecular form of substrate
- Non-hydrolytic removal of a group from a substrate
- Joining of two molecules by the formation of new bonds
- Hydrolysis of substrate

Click to Save Answer

Question # 6 of 6 (Start time: 10:24:46 PM, 23 August 2022)

Removal of phosphoryl groups is catalyzed by_ _ _ _ _ _ _

Select the correct option

- Dinitrogenase reductase
- Protein phosphatases
- Diphteria toxin and cholera toxin
- Protein kinases

Click to Sa

BIO303 - Biochemistry II (2nd quiz)

Question # 8 of 8 (Start time: 10:27:06 PM, 23 August 2022)

The condensation reaction is catalysed by _____ enzyme

Select the correct option

- Isomerase
- Fumerase
- Ligase
- Reductase



Click to

BIO303 - Biochemistry II (2nd quiz) **Quiz Start Time: 11:45 PM, 23**

Question # 1 of 8 (Start time: 11:45:37 PM, 23 August 2022)

Which one of the following is a lipid hydrolyzing enzyme?

Select the correct option

- ○ Peptidase
- ○ Lyase
- ◉ Hydrolase
- ○ Phospholipase

Click to Save Answer & Move to Next

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 11:45 PM, 23 August 2022

Question # 2 of 8 (Start time: 11:46:20 PM, 23 August 2022)

Total Marks: 1

A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called a _____

Select the correct option

- ○ Cofactor
- ○ Coenzyme
- ◉ Prosthetic group
- ○ Ligand

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 11:45 PM, 23 August 2022

Question # 3 of 8 (Start time: 11:47:14 PM, 23 August 2022)

Total Marks: 1

In a chemical reaction the position and direction of equilibrium are not affected by _____

Select the correct option

- ○ Substrate
- ○ Product
- ○ ES complex
- ◉ Catalyst

Click to Save Answer & Move to Next Question

Question # 5 of 8 (Start time: 11:48:01 PM, 23 August 2022)

Total Marks: 1

The free energy of the ground state of P is _____ than that of S

Select the correct option

- ○ Higher
- ○ Equal
- ◉ lower
- ○ Different

Click to Save Answer & Move to Next Question

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 11:45 PM, 23 August 2022

Question # 7 of 8 (Start time: 11:48:34 PM, 23 August 2022)

Total Marks: 1

Removal of phosphoryl groups is catalyzed by_____

Select the correct option

- ○ Diphteria toxin and cholera toxin
- ○ Protein kinases
- ◉ Protein phosphatases
- ○ Dinitrogenase reductase

Click to Save Answer & Move to Next Question

sec(s)

BIO303 - Biochemistry II (2nd quiz)

Quiz Start Time: 11:45 PM, 23 August 2022

Question # 8 of 8 (Start time: 11:48:48 PM, 23 August 2022)

Total Marks: 1

The substrate for amylase is

Select the correct option

○	Cane sugar
◉	Starch
○	Ribose
○	Lactose

Click to Save Answer & Move to Next Question